

Nazarbayev University
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Fall 2019

Problem Set for Tutorial 1

To be discussed on T1, August 23, 2019

1. You are taking two courses this semester, Economics and Psychology. You have quizzes coming up in both classes. The table below shows your grade on each quiz for different numbers of hours studying for each quiz. For instance, the second row implies that one hour of studying for Psychology will generate an expected grade of 56 on Psychology, whereas one hour of studying for Economics will generate an expected grade of 63 on Economics.

Hours of study	Psychology	Economics
0	50	56
1	56	63
2	60	68
3	62	71

Your goal is to maximize your average grade on the two quizzes. . Use the idea of optimization in differences to decide how much time you would spend studying for each quiz if you had a total of:

- (a) 1 hour to prepare.
 - (b) 2 hours to prepare.
 - (c) 3 hours to prepare.
2. The demand for computers is

$$q_D = 13 - 2p$$

where p is the price of computers. Initially, the supply of computers is $q_S = 4 + p$

- (a) Carefully plot the demand and supply schedules. Your diagram should be correctly labeled and indicate all important points (intercepts and slope).
- (b) Calculate the equilibrium price and quantity (q^* , p^*) and indicate them in the diagram above.

Now suppose that the prices of memory chips and motherboards, which are two important components in computers, rise and as a consequence, the supply curve for computers becomes

$$q_S = p.$$

- (c) In the demand and supply diagram from the previous item, show the effect of the increase in the prices of memory chips and motherboards.
 - (d) Show the effect of the increase in the prices of memory chips and motherboards on the equilibrium price and quantity by calculating their new values and indicate them in the previous diagram.
3. Suppose that an economy has two individuals, Alejandro and Oleg. Alejandro's demand curve is given by $q_D^A = 100 - p$ and Oleg's demand curve is given by $q_D^O = 20 - 0.5P$

- (a) Perform horizontal summation of these demand curves to find the market demand curve. Denote this curve by Q_D
- (b) Suppose that the supply curve is given by $Q_S = 5 + p$. What is the equilibrium market quantity and price?
- (c) Now suppose that the supply curve is given by $Q_S = 5 + 0.5p$. Compute the equilibrium market quantity and price in this case. Why is it different from the previous item?
- (d) Illustrate all the previous items in a correctly labeled diagram.